Elementary Analysis

Chapter 2, Section 12

Limit Superior and Limit Inferior
Exercises 12.1–12.14, with every subpart

Mathjock

Text and numbering. Kenneth A. Ross, *Elementary Analysis: The Theory of Calculus*, second edition (Springer, 2013), Chapter 2. Exercise numbers follow the copy already linked on Mathjock. These are independently written worked solutions, not the publisher's solutions manual. Each exercise is identified by a short mathematical description.

Reading philosophy. Every argument states its goal, records the assumptions being used, and explains the intermediate algebra or logical inference. Definitions, sign conditions, and both directions of an "if and only if" are made explicit. A cited rule is named at the point where it is used.

Conventions. $\mathbb{N} = \{1, 2, 3, \ldots\}$, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, and $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z},\ q \neq 0\}$. Unless a different domain is stated, sequence indices run through $\mathbb{N}$ and all terms and finite limits are real. The empty sum is 0, and $0! = 1$.

Contents

12 Limit superior and limit inferior **3**

12.1 Exercise 12.1: Eventual comparison passes to both limiting bounds 3

12.2 Exercise 12.2: A zero limit superior of absolute values 4

12.3 Exercise 12.3: Two synchronized sequences of period four 5

12.4 Exercise 12.4: The limit superior of a sum is at most the sum of the limit superiors 6

12.5 Exercise 12.5: The corresponding lower bound for the limit inferior of a sum 7

12.6 Exercise 12.6: Scaling the upper and lower limiting bounds 8

12.7 Exercise 12.7: A positive multiplier preserves an infinite limit superior 10

12.8 Exercise 12.8: A product bound for bounded nonnegative sequences 10

12.9 Exercise 12.9: A positive lower bound and products tending upward 11

12.10 Exercise 12.10: Detecting boundedness with the limit superior of absolute values . . 12

12.11 Exercise 12.11: The lower ratio–root inequality 13

12.12 Exercise 12.12: Arithmetic means and the limits of the original sequence 14

12.13 Exercise 12.13: Limit inferior and superior as eventual bounds 16

12.14 Exercise 12.14: The growth of the geometric mean of the first integers 17

12 Limit superior and limit inferior

For each integer $N \geq 0$, the tail after N is the nonempty set

$$E_N = \{s_n : n > N\}.$$

Definition 10.6 gives

$$\limsup s_n = \lim_{N\to\infty} \sup E_N, \qquad \liminf s_n = \lim_{N\to\infty} \inf E_N.$$

These limits may be infinite. When a tail bound is infinite, we check the case through the definitions and extended order before applying any theorem about sequences of real numbers.

A tail supremum need not be attained by a term. Likewise, a term may lie above the lim sup or below the lim inf at arbitrarily large indices. The solutions check the relevant bound and eventuality conditions explicitly.

Tail notation and the meaning of infinite bounds

For a real sequence (x_n) and an integer $N \geq 1$, write

$$X_N = \{x_n : n > N\}, \qquad u_N(x) = \inf X_N, \qquad v_N(x) = \sup X_N.$$

The set X_N is nonempty because it contains x_{N+1}. We use the extended-bound conventions of Section 5: $u_N(x) = -\infty$ if the tail is unbounded below, and $v_N(x) = +\infty$ if it is unbounded above. For a nonempty set of real terms, a tail infimum cannot be $+\infty$, and a tail supremum cannot be $-\infty$. By Definition 10.6,

$$\liminf_{n\to\infty} x_n = \lim_{N\to\infty} u_N(x), \qquad \limsup_{n\to\infty} x_n = \lim_{N\to\infty} v_N(x).$$

Removing more terms enlarges the infimum and decreases the supremum, since $X_{N+1} \subseteq X_N$.

We will also use a finite-prefix observation. If a tail of a real sequence has an upper bound U, then adjoining its finitely many omitted terms still gives a bounded-above set: take the maximum of U and those finitely many terms. The corresponding minimum proves the analogous statement for lower bounds. Thus a sequence that is unbounded above, or unbounded below, has the same property on every tail. Infinite symbols will be handled through order and these conventions, rather than through real-number algebra.

12.1 Exercise 12.1: Eventual comparison passes to both limiting bounds

What we must show. Suppose there is an N_0 such that $s_n \leq t_n$ for all $n > N_0$. Prove

$$\liminf s_n \leq \liminf t_n, \qquad \limsup s_n \leq \limsup t_n.$$

No boundedness assumption is imposed on the two sequences.

Step 1: make the eventual comparison valid on the tails under discussion.

Enlarge N_0, if necessary, to a positive integer. This keeps the hypothesis true, because it discards additional initial terms. Fix any integer $N \geq N_0$. For every $n > N$ we then have $s_n \leq t_n$.

For the upper tail bounds, if $v_N(t) = +\infty$, the extended-order comparison $v_N(s) \le v_N(t)$ holds immediately. Otherwise $v_N(t)$ is real, and

$$s_n \le t_n \le v_N(t) \qquad (n > N).$$

Thus $v_N(t)$ is an upper bound of the tail of (s_n), so its least upper bound satisfies $v_N(s) \le v_N(t)$.

For the lower tail bounds, if $u_N(s) = -\infty$, the comparison $u_N(s) \le u_N(t)$ is automatic in the extended order. Otherwise $u_N(s)$ is real, and

$$u_N(s) \le s_n \le t_n \qquad (n > N).$$

Thus $u_N(s)$ is a lower bound of the tail of (t_n), so it is at most that tail's greatest lower bound. We have proved

$$u_N(s) \le u_N(t), \qquad v_N(s) \le v_N(t) \qquad (N \ge N_0).$$

Step 2: pass to the lower limiting bounds, checking infinite cases.

If (s_n) is unbounded below, every tail infimum $u_N(s)$ equals $-\infty$. Therefore $\liminf s_n = -\infty \le \liminf t_n$, and there is nothing further to prove for the lower bounds in this case.

If (s_n) is bounded below, choose a real lower bound a. For $n > N_0$, the hypothesis gives $t_n \ge s_n \ge a$. The finite-prefix observation shows that (t_n) is bounded below too. Consequently both $u_N(s)$ and $u_N(t)$ are real for every N. Their limits exist in the extended sense by Definition 10.6. Apply Exercise 9.9(c), the order rule for existing limits of real-valued sequences, to $u_N(s) \le u_N(t)$:

$$\lim_{N\to\infty} u_N(s) \le \lim_{N\to\infty} u_N(t).$$

The limits may be $+\infty$, but the terms to which the exercise was applied are real. By definition this is $\liminf s_n \le \liminf t_n$.

Step 3: pass to the upper limiting bounds, checking infinite cases.

If (t_n) is unbounded above, then every $v_N(t) = +\infty$, so $\limsup s_n \le +\infty = \limsup t_n$. If (t_n) is bounded above by a real number b, then $s_n \le t_n \le b$ for $n > N_0$. The finite-prefix observation shows that (s_n) is bounded above as well. Hence $v_N(s)$ and $v_N(t)$ are real-valued sequences, with existing extended limits. Applying Exercise 9.9(c) to their eventual inequality gives

$$\lim_{N\to\infty} v_N(s) \le \lim_{N\to\infty} v_N(t),$$

which is $\limsup s_n \le \limsup t_n$.

Conclusion. Both comparisons hold:

$$\boxed{\liminf s_n \le \liminf t_n, \qquad \limsup s_n \le \limsup t_n}.$$

The proof includes infinite limiting bounds. It does not treat $+\infty$ or $-\infty$ as individual real terms in an application of a real-sequence limit rule.

12.2 Exercise 12.2: A zero limit superior of absolute values

What we must show. Prove the equivalence

$$\limsup |s_n| = 0 \quad \Longleftrightarrow \quad s_n \longrightarrow 0.$$

Forward direction, Step 1: derive the boundedness needed for finite tail suprema.

Assume $\limsup |s_n| = 0$ and put $a_n = |s_n| \geq 0$. If (a_n) were unbounded above, every tail would be unbounded above by the finite-prefix observation. Its every tail supremum would therefore be $+\infty$, giving $\limsup a_n = +\infty$, contrary to the assumed value 0. Thus (a_n) is bounded above. Because all $a_n \geq 0$, it is bounded below as well.

It follows that the numbers

$$V_N = \sup\{|s_n| : n > N\}$$

are finite and nonnegative for every positive integer N. Their nonnegativity follows, for example, by comparing V_N with the member $|s_{N+1}| \geq 0$ of the tail. The hypothesis and Definition 10.6 now say $V_N \to 0$.

Forward direction, Step 2: control every term in one sufficiently late tail.

Let $\varepsilon > 0$. From $V_N \to 0$, choose a positive integer N_* large enough that $|V_{N_*} - 0| < \varepsilon$. Since $V_{N_*} \geq 0$, this means $V_{N_*} < \varepsilon$. For every integer $n > N_*$, the definition of tail supremum gives

$$|s_n - 0| = |s_n| \leq V_{N_*} < \varepsilon.$$

This is Definition 7.1 for $s_n \to 0$. Boundedness was obtained from the hypothesis before any finite-tail estimate was used.

Reverse direction, Step 1: verify convergence of the absolute values.

Assume $s_n \to 0$. Let $\varepsilon > 0$. The definition gives a threshold N such that $n > N$ implies $|s_n - 0| < \varepsilon$. Since $|s_n| \geq 0$, the error in the sequence of absolute values is exactly

$$||s_n| - 0| = |s_n| = |s_n - 0| < \varepsilon.$$

Thus $|s_n| \to 0$ directly from the definition.

Reverse direction, Step 2: identify its limit superior.

Theorem 10.7 says that when a sequence has a limit, its limit inferior and limit superior equal that limit. Applying it to the convergent sequence $(|s_n|)$ gives $\limsup |s_n| = 0$.

Conclusion. The two implications prove $\boxed{\limsup |s_n| = 0 \iff s_n \to 0}$.

12.3 Exercise 12.3: Two synchronized sequences of period four

What we must show. Compute all seven requested combinations for the sequences whose repeated blocks are $(0, 1, 2, 1)$ and $(2, 1, 1, 0)$.

Step 1: keep the phases aligned in one residue table.

The index n has one of four residues modulo 4. Reading the entries at the same index gives

Residue of n	s_n	t_n	$s_n + t_n$	$s_n t_n$
1	0	2	2	0
2	1	1	2	1
3	2	1	3	2
0	1	0	1	0

For instance, at indices $n = 4j + 3$ the entries are $s_n = 2$ and $t_n = 1$, so the sum is 3 and the product is 2. Every row occurs at arbitrarily large indices: by taking the integer j sufficiently large, its index exceeds any prescribed threshold. For the residue 0, use indices $4, 8, 12, \ldots$.

Step 2: identify each tail's minimum and maximum.

Every tail of (s_n) contains the values $0, 1, 2$ and no others. Thus its infimum is 0 and its supremum is 2. The same is true for every tail of (t_n). The sum has the repeated block $(2, 2, 3, 1)$, so every tail of the sum contains exactly the values $1, 2, 3$; its infimum is 1 and its supremum is 3. The product has the repeated block $(0, 1, 2, 0)$, so its tail infima are 0 and its tail suprema are 2. Taking the limits of these constant sequences of tail bounds gives

$$\liminf s_n = 0, \quad \limsup s_n = 2, \qquad \liminf t_n = 0, \quad \limsup t_n = 2,$$

$$\liminf(s_n + t_n) = 1, \quad \limsup(s_n + t_n) = 3, \qquad \liminf(s_n t_n) = 0, \quad \limsup(s_n t_n) = 2.$$

Step 3: evaluate every requested expression.

(a) $\liminf s_n + \liminf t_n = 0 + 0 = \boxed{0}$,

(b) $\liminf(s_n + t_n) = \boxed{1}$,

(c) $\liminf s_n + \limsup t_n = 0 + 2 = \boxed{2}$,

(d) $\limsup(s_n + t_n) = \boxed{3}$,

(e) $\limsup s_n + \limsup t_n = 2 + 2 = \boxed{4}$,

(f) $\liminf(s_n t_n) = \boxed{0}$,

(g) $\limsup(s_n t_n) = \boxed{2}$.

Conclusion. The phase alignment matters. The largest value of (s_n) occurs when $t_n = 1$, whereas the largest value of (t_n) occurs when $s_n = 0$. Adding the separate maxima therefore gives 4, even though the paired sum has maximum 3.

12.4 Exercise 12.4: The limit superior of a sum is at most the sum of the limit superiors

What we must show. For bounded real sequences (s_n) and (t_n), prove

$$\limsup(s_n + t_n) \le \limsup s_n + \limsup t_n.$$

Step 1: establish that all relevant tail bounds are finite.

Choose real constants $A, B \ge 0$ with $|s_n| \le A$ and $|t_n| \le B$ for every n. The triangle inequality gives $|s_n + t_n| \le A + B$, so the sum sequence is bounded as well. For every positive integer N, the three tails are nonempty and bounded, and hence the numbers

$$S_N = \sup\{s_n : n > N\}, \quad T_N = \sup\{t_n : n > N\}, \quad W_N = \sup\{s_n + t_n : n > N\}$$

are real. Their limits as $N \to \infty$ are also finite, since the underlying sequences are bounded.

Step 2: prove the finite-tail inequality.

For every integer $n > N$, the upper-bound properties give $s_n \le S_N$ and $t_n \le T_N$. Adding yields

$$s_n + t_n \le S_N + T_N.$$

Thus the real number $S_N + T_N$ is an upper bound of the paired-sum tail. Its least upper bound satisfies

$$W_N \le S_N + T_N.$$

This argument does not require S_N or T_N to be attained. It also does not require a single index where both separate bounds are approached at once.

Step 3: take limits of the real tail-bound sequences.

By Definition 10.6,

$$S_N \to \limsup s_n, \qquad T_N \to \limsup t_n, \qquad W_N \to \limsup(s_n + t_n).$$

The sum law from Section 9 gives

$$S_N + T_N \longrightarrow \limsup s_n + \limsup t_n.$$

Apply Exercise 9.9(c) to the termwise comparison $W_N \le S_N + T_N$. The resulting comparison of limits is

$$\limsup(s_n + t_n) \le \limsup s_n + \limsup t_n.$$

Conclusion. This proves the desired inequality. The boundedness hypotheses ensure that every sum of limiting bounds used here is a sum of real numbers; no undefined combination of infinite symbols has been introduced.

12.5 Exercise 12.5: The corresponding lower bound for the limit inferior of a sum

What we must show. Use Exercises 11.8 and 12.4 to prove, for bounded real sequences,

$$\liminf(s_n + t_n) \ge \liminf s_n + \liminf t_n.$$

Step 1: verify that the reflected sequences meet the hypotheses.

The sequences $(-s_n)$ and $(-t_n)$ are bounded because $|-s_n| = |s_n|$ and $|-t_n| = |t_n|$. Their sum is $-s_n - t_n = -(s_n + t_n)$, which is bounded too. Exercise 12.4 therefore applies and gives

$$\limsup\big(-(s_n + t_n)\big) \le \limsup(-s_n) + \limsup(-t_n).$$

Step 2: replace each reflected upper limiting bound.

Exercise 11.8 states $\liminf x_n = -\limsup(-x_n)$. For the bounded sequences here, all these values are real. Taking their negatives also gives $\limsup(-x_n) = -\liminf x_n$. Substitute this identity in the three places in Step 1:

$$-\liminf(s_n + t_n) \le -\liminf s_n - \liminf t_n = -\big(\liminf s_n + \liminf t_n\big).$$

Step 3: reverse the inequality by negation.

Multiplying a real inequality by -1 reverses its direction. Negating both sides of the last comparison, and taking the negative twice in each expression, yields

$$\liminf(s_n + t_n) \geq \liminf s_n + \liminf t_n.$$

Conclusion. The claimed lower inequality follows. The sign reversal is the reason that reflecting the upper inequality from Exercise 12.4 produces a lower inequality here.

12.6 Exercise 12.6: Scaling the upper and lower limiting bounds

What we must show. For bounded (s_n) and $k \geq 0$, prove (a) $\limsup(ks_n) = k \limsup s_n$ and (b) $\liminf(ks_n) = k \liminf s_n$. Determine the correct formulas in (c) when $k < 0$.

Preliminary step: scaling preserves boundedness.

Choose $A \geq 0$ with $|s_n| \leq A$ for every n. For any fixed real k, the identity $|ks_n| = |k|\,|s_n| \leq |k|\,A$ shows that (ks_n) is bounded. Its tails are nonempty. Hence all tail bounds for (s_n) and (ks_n) below are real numbers. Write

$$m_N = \inf\{s_n : n > N\}, \qquad M_N = \sup\{s_n : n > N\}.$$

The limits of these two sequences are $\liminf s_n$ and $\limsup s_n$, respectively, and are finite by boundedness.

Parts (a) and (b), the case $k = 0$.

If $k = 0$, every scaled term is zero. Every scaled tail is the singleton $\{0\}$, whose infimum and supremum are zero. Therefore both limiting bounds of (ks_n) are zero. The expressions $0 \cdot \liminf s_n$ and $0 \cdot \limsup s_n$ are also zero, because their other factors are finite real numbers. Thus both requested formulas hold in this case.

Part (a), Step 1: prove the upper-tail scaling formula for $k > 0$.

Fix $k > 0$ and an integer $N \geq 1$. Since $s_n \leq M_N$ for all $n > N$, multiplication by k gives $ks_n \leq kM_N$. Thus kM_N is an upper bound of the scaled tail.

To check that it is the least upper bound, let U be any real upper bound of that tail. Then $ks_n \leq U$ for every $n > N$. Division by $k > 0$ gives $s_n \leq U/k$ for all these indices, so U/k is an upper bound of the original tail. The least-upper-bound property gives $M_N \leq U/k$. Multiplying by $k > 0$ gives $kM_N \leq U$. We have proved both defining properties of a supremum, so

$$\sup\{ks_n : n > N\} = kM_N.$$

Part (a), Step 2: take the limit of the real tail bounds.

By the constant-multiple limit law from Section 9,

$$kM_N \longrightarrow k \limsup s_n.$$

Definition 10.6 identifies the limit of the left side of the tail identity as $\limsup(ks_n)$. Hence

$$\boxed{\limsup(ks_n) = k \limsup s_n} \qquad (k > 0).$$

Together with the zero case, this proves part (a) for all $k \geq 0$.

Part (b), Step 1: prove the lower-tail scaling formula for $k > 0$.

For every $n > N$, $m_N \leq s_n$; multiplying by $k > 0$ gives $km_N \leq ks_n$. Thus km_N is a lower bound of the scaled tail. If L is any real lower bound of that tail, then $L \leq ks_n$ for every $n > N$. Divide by $k > 0$ to obtain $L/k \leq s_n$. Consequently L/k is a lower bound of the original tail. Since m_N is its greatest lower bound, $L/k \leq m_N$, and hence $L \leq km_N$. These two properties establish

$$\inf\{ks_n : n > N\} = km_N.$$

Taking limits of this equality and using the constant-multiple law gives

$$\boxed{\liminf(ks_n) = k \liminf s_n} \qquad (k > 0).$$

The zero case already proved extends it to all $k \geq 0$.

Part (b), Step 2: relate the formula to the reflection hint.

For completeness, the same lower formula follows from part (a) and Exercise 11.8. When $k > 0$,

$$\begin{aligned} \liminf(ks_n) &= -\limsup(-ks_n), \\ &= -\limsup(k(-s_n)), \\ &= -k\limsup(-s_n), \\ &= k\liminf s_n. \end{aligned}$$

The third equality uses part (a) for the bounded sequence $(-s_n)$; the first and last use reflection. All negatives here are negatives of finite real numbers.

Part (c), Step 1: a negative multiplier exchanges the tail bounds.

Now let $k < 0$ and fix N. Multiplication by k reverses order. Since $m_N \leq s_n$ for every $n > N$, we have $ks_n \leq km_N$; thus km_N is an upper bound of the scaled tail. For any other upper bound U, the inequalities $ks_n \leq U$ become $s_n \geq U/k$ after division by $k < 0$. Hence U/k is a lower bound of the original tail, so $U/k \leq m_N$. Multiplying this last inequality by $k < 0$ reverses it and gives $U \geq km_N$. Therefore

$$\sup\{ks_n : n > N\} = km_N \qquad (k < 0).$$

Similarly, $s_n \leq M_N$ gives $ks_n \geq kM_N$, so kM_N is a lower bound of the scaled tail. If L is any other lower bound, $L \leq ks_n$ gives $L/k \geq s_n$ after division by $k < 0$. Thus L/k is an upper bound of the original tail and $M_N \leq L/k$. Multiplying by $k < 0$ gives $kM_N \geq L$. It follows that

$$\inf\{ks_n : n > N\} = kM_N \qquad (k < 0).$$

Part (c), Step 2: take finite limits.

The constant-multiple law, now with a fixed negative constant, still applies to these real convergent tail-bound sequences. Taking limits of the two identities gives

$$\boxed{\limsup(ks_n) = k\liminf s_n, \qquad \liminf(ks_n) = k\limsup s_n} \qquad (k < 0).$$

Conclusion. A positive multiplier preserves the upper and lower limiting bounds separately. A zero multiplier makes both zero. A negative multiplier exchanges their roles, as the two formulas in part (c) state. All statements were proved without assuming that any tail supremum or infimum is attained.

12.7 Exercise 12.7: A positive multiplier preserves an infinite limit superior

What we must show. Suppose $\limsup s_n = +\infty$ and $k > 0$. Prove $\limsup(ks_n) = +\infty$ without assuming that (s_n) is bounded.

Step 1: prove that every tail of the original sequence is unbounded above.

Suppose some tail were bounded above by a real U. Then $s_n \le U$ for every index in that tail, so the eventual comparison result of Exercise 12.1, applied to (s_n) and the constant sequence U, would give

$$\limsup s_n \le \limsup U = U.$$

This contradicts $\limsup s_n = +\infty$. Thus no such tail bound exists: for each integer N and each real R, some index $n > N$ satisfies $s_n > R$.

Step 2: prove that every scaled tail is unbounded above.

Fix any positive integer N and any proposed real upper bound M for the scaled tail. Since $k > 0$, the quotient M/k is real. Step 1, with $R = M/k$, gives an index $n > N$ such that $s_n > M/k$. Multiplying by the positive number k preserves strict inequality:

$$ks_n > M.$$

Thus M is not an upper bound. Because M was arbitrary, the scaled tail is unbounded above. Because N was arbitrary, every scaled tail has this property.

Step 3: apply the definition of the infinite tail bound.

For every N, the extended-bound convention gives

$$\sup\{ks_n : n > N\} = +\infty.$$

Its limiting value is therefore $+\infty$, which is exactly $\limsup(ks_n)$ by Definition 10.6.

Conclusion. We obtain $\boxed{\limsup(ks_n) = +\infty}$. The proof divided and multiplied only real numbers, such as M and M/k, and used unboundedness to interpret the infinite result.

12.8 Exercise 12.8: A product bound for bounded nonnegative sequences

What we must show. Let (s_n) and (t_n) be bounded sequences with $s_n \ge 0$ and $t_n \ge 0$ for every n. Prove

$$\limsup(s_n t_n) \le (\limsup s_n)(\limsup t_n).$$

Step 1: give finite bounds for the three sequences.

Choose $A, B \ge 0$ such that $0 \le s_n \le A$ and $0 \le t_n \le B$ for every n. Because $t_n \ge 0$, multiplication of $s_n \le A$ by t_n gives $s_n t_n \le A t_n$. Because $A \ge 0$, multiplication of $t_n \le B$ by A gives $A t_n \le AB$. Thus

$$0 \le s_n t_n \le A t_n \le AB.$$

The product sequence is bounded and nonnegative too. For every N, all three tails are nonempty and bounded, and their suprema

$$S_N = \sup\{s_n : n > N\}, \quad T_N = \sup\{t_n : n > N\}, \quad P_N = \sup\{s_n t_n : n > N\}$$

are finite and nonnegative. For example, $0 \le s_{N+1} \le S_N$, which proves $S_N \ge 0$; the other two nonnegativity statements follow in the same way from their tails.

Step 2: prove the paired-product tail inequality.

For each $n > N$, the tail upper bounds give $s_n \le S_N$ and $t_n \le T_N$. Multiply the first inequality by $t_n \ge 0$:

$$s_n t_n \le S_N t_n.$$

Multiply the second by $S_N \ge 0$:

$$S_N t_n \le S_N T_N.$$

The sign checks are necessary for both comparisons. Combining them shows $s_n t_n \le S_N T_N$ for every $n > N$. Hence $S_N T_N$ is an upper bound of the product tail. Its least upper bound satisfies

$$P_N \le S_N T_N.$$

Neither separate tail supremum needs to be attained, and no common index attaining both was selected.

Step 3: compare the finite limits of the tail-bound sequences.

Definition 10.6 gives

$$S_N \to \limsup s_n, \qquad T_N \to \limsup t_n, \qquad P_N \to \limsup(s_n t_n).$$

All limits here are finite because the original sequences and the product sequence are bounded. The product limit law from Section 9 therefore yields

$$S_N T_N \longrightarrow (\limsup s_n)(\limsup t_n).$$

Applying Exercise 9.9(c) to $P_N \le S_N T_N$ gives the requested comparison of limiting values.

Conclusion. We have proved

$$\boxed{\limsup(s_n t_n) \le (\limsup s_n)(\limsup t_n)}.$$

The proof includes the case where either upper limiting bound is zero, because no division by one of those bounds was used.

12.9 Exercise 12.9: A positive lower bound and products tending upward

What we must show. Prove the product assertions in parts (a) and (b), and identify Exercise 12.7 as the special case requested in part (c).

A preliminary fact: extracting one positive lower bound.

Write $\alpha = \liminf t_n$, and suppose $\alpha > 0$. The value $\alpha = +\infty$ is allowed. We claim that there are a real number $c > 0$ and an integer N_0 such that

$$t_n \ge c \qquad (n > N_0).$$

If α is finite, choose $c = \alpha/2$; then $0 < c < \alpha$. If $\alpha = +\infty$, choose $c = 1$ instead. In either case, the tail infima $\inf\{t_n : n > N\}$ eventually exceed c, by their convergence to α. Each term in such a tail is

at least its infimum, and therefore exceeds c. Replacing this strict inequality by $\geq c$ gives the claim. This argument also covers an initially unbounded sequence: since $\liminf t_n > 0$, some tail has a finite lower bound.

(a) An ordinary limit of $+\infty$.

Assume $s_n \to +\infty$. To prove $s_n t_n \to +\infty$, fix an arbitrary real threshold $M > 0$. By the definition of this limit, there is N_1 such that

$$n > N_1 \quad \Longrightarrow \quad s_n > M/c.$$

Since $M/c > 0$, these s_n are positive. If $n > \max\{N_0, N_1\}$, we can multiply $t_n \geq c$ by the positive number s_n, obtaining

$$s_n t_n \geq c s_n > c(M/c) = M.$$

Thus every sufficiently late product exceeds M. Checking positive thresholds suffices: exceeding any positive number also exceeds all smaller, possibly negative, thresholds. Hence $s_n t_n \to +\infty$.

(b) A limit superior of $+\infty$.

Now assume only $\limsup s_n = +\infty$. Every tail of (s_n) is unbounded above. Indeed, a tail bounded above, together with the finitely many preceding real terms, would bound the whole sequence above and force its limit superior to be at most a real upper bound.

Fix a tail index N and a positive threshold M. The unbounded tail beyond $\max\{N, N_0\}$ contains an index n for which $s_n > M/c$. At that index $n > N$, and the same sign calculation as in part (a) gives $s_n t_n > M$. Thus every tail of the product sequence is unbounded above. Its tail suprema are all $+\infty$, so its limit superior is $+\infty$.

The distinction between (a) and (b) is in the quantifiers: part (a) puts every sufficiently late product above M; part (b) produces at least one such product in every tail.

(c) Recover Exercise 12.7.

If $t_n = k > 0$ for every n, then $\liminf t_n = k > 0$ and $s_n t_n = k s_n$. Part (b) therefore states exactly that

$$\limsup s_n = +\infty \quad \Longrightarrow \quad \limsup(k s_n) = +\infty \qquad (k > 0).$$

Conclusion. Both product conclusions hold; a positive constant second factor gives Exercise 12.7.

12.10 Exercise 12.10: Detecting boundedness with the limit superior of absolute values

What we must show. Prove both directions of

$$(s_n) \text{ is bounded} \quad \Longleftrightarrow \quad \limsup |s_n| < +\infty.$$

Plan. Absolute values express upper and lower bounds on s_n in one nonnegative sequence. An eventual bound becomes a bound on all terms after we include the finite initial segment.

A bounded sequence gives a finite limit superior.

If (s_n) is bounded, there is a finite $B \geq 0$ with $|s_n| \leq B$ for every n. Thus for every N,

$$0 \leq \sup\{|s_n| : n > N\} \leq B.$$

These tail suprema decrease as N increases. The monotone sequence theorem gives a limit in $[0, B]$, and by definition that limit is $\limsup |s_n|$. In particular it is less than $+\infty$.

A finite limit superior gives an eventual bound.

Conversely, suppose $L = \limsup |s_n| < +\infty$. Since all the absolute values are nonnegative, $L \geq 0$, so L is a real number. The sequence $(|s_n|)$ cannot be unbounded above: if it were, every tail would be unbounded above, since deleting finitely many real terms cannot remove unboundedness. All its tail suprema would then be $+\infty$, contradicting the finite value of L.

Equivalently, some tail has a finite supremum. Once that happens, all subsequent tail suprema are finite and converge to L. Their convergence yields an integer $N \geq 1$ with

$$|s_n| \leq \sup\{|s_j| : j > N\} < L + 1 \qquad (n > N).$$

Include the early terms.

The finite set of real numbers $\{|s_1|, \ldots, |s_N|, L+1\}$ has a finite maximum B. If $n \leq N$, the definition of B gives $|s_n| \leq B$. If $n > N$, the tail estimate gives $|s_n| < L + 1 \leq B$. Consequently $|s_n| \leq B$ for every index n.

Conclusion. The two conditions are equivalent. The limit superior bounds the tail, and a finite maximum absorbs the initial terms.

12.11 Exercise 12.11: The lower ratio–root inequality

What we must show. For a sequence with $s_n \neq 0$ at every index, prove the first inequality of Theorem 12.2:

$$\liminf \left| \frac{s_{n+1}}{s_n} \right| \;\leq\; \liminf |s_n|^{1/n}.$$

Plan. A lower bound on successive ratios becomes a geometric lower bound on the terms. Taking nth roots removes the fixed factor.

Name the two nonnegative extended limits.

Set

$$A = \liminf \left| \frac{s_{n+1}}{s_n} \right|, \qquad B = \liminf |s_n|^{1/n}.$$

Every ratio is well defined and positive because no s_n is zero. The roots are positive as well. Therefore $A, B \in [0, +\infty]$. If $A = 0$, then $A \leq B$ follows immediately from $B \geq 0$.

An arbitrary positive number below A is an eventual ratio bound.

Suppose $A > 0$, and choose any real a with $0 < a < A$. When $A = +\infty$, this means that a can be any positive real number. The definition of the limit inferior gives an integer $K \geq 1$ such that

$$\left| \frac{s_{j+1}}{s_j} \right| > a \qquad (j \geq K).$$

To see this directly, some tail infimum of these ratios exceeds a; each ratio in that tail is at least the infimum. Increase the starting index to K if needed.

Multiply the consecutive inequalities.

For $n > K$, cancellation of the nonzero terms gives

$$|s_n| = |s_K| \prod_{j=K}^{n-1} \left| \frac{s_{j+1}}{s_j} \right| \geq |s_K|\, a^{n-K}.$$

The weak inequality also holds when $n = K$, since the product then has no factors and equals 1. Define the fixed positive constant $C = |s_K|\, a^{-K}$. The estimate becomes

$$|s_n| \geq Ca^n \qquad (n \geq K).$$

The map $x \mapsto x^{1/n}$ is increasing on nonnegative reals, so

$$|s_n|^{1/n} \geq aC^{1/n} \qquad (n \geq K).$$

Pass to the limit inferior and then let a approach A.

Since $C > 0$, the elementary root limit gives $C^{1/n} \to 1$. Thus $aC^{1/n} \to a$, and eventual order (Exercise 12.1) yields $B \geq a$.

If A is finite and $B < A$, then B is finite and a number a strictly between B and A satisfies $a > 0$. This contradicts $B \geq a$. Hence $B \geq A$ in the finite case. If $A = +\infty$, the inequality $B \geq a$ holds for every positive real a; a finite B would be contradicted by taking $a > B$. Therefore $B = +\infty$.

Conclusion. In every case the lower ratio limit inferior is at most the root limit inferior. This proof does not rely on Corollary 12.3, whose proof uses the inequality being established here.

12.12 Exercise 12.12: Arithmetic means and the limits of the original sequence

What we must show. For $s_n \geq 0$, put

$$\sigma_n = \frac{s_1 + \cdots + s_n}{n}.$$

Prove the four-term inequality in part (a), deduce part (b), and give the counterexample requested in part (c).

(a), preparation: name the bounds and check nonnegativity.

Write $\alpha = \liminf s_n$ and $\beta = \limsup s_n$. Because $s_n \geq 0$, all averages are nonnegative, and $\alpha, \beta, \liminf \sigma_n, \limsup \sigma_n$ belong to $[0, +\infty]$. The inequality $\liminf \sigma_n \leq \limsup \sigma_n$ holds for every sequence by Theorem 10.7. We prove the two outside inequalities separately.

(a), upper bound: verify the estimate in the hint.

First suppose $\beta < +\infty$. The sequence (s_n) must then be bounded above: otherwise every tail would be unbounded above, and its limit superior would be $+\infty$. Set

$$U_N = \sup\{s_n : n > N\}, \qquad P_N = s_1 + \cdots + s_N.$$

These are finite, with $U_N, P_N \geq 0$. If $n > N$, each of the $n - N$ remaining terms is at most U_N, so

$$\sigma_n \leq \frac{P_N}{n} + \frac{n-N}{n} U_N \leq \frac{P_N}{n} + U_N.$$

The second inequality uses $U_N \geq 0$ and $(n-N)/n \leq 1$. If $n > M > N$, the inequality $P_N/n \leq P_N/M$ uses $P_N \geq 0$. Taking the supremum over $n > M$ gives the hint exactly:

$$\sup\{\sigma_n : n > M\} \leq \frac{P_N}{M} + U_N.$$

Hold N fixed and let $M \to \infty$. The left side tends to $\limsup \sigma_n$, while $P_N/M \to 0$, yielding $\limsup \sigma_n \leq U_N$. This holds for every N. Letting $N \to \infty$ and using $U_N \to \beta$, we get $\limsup \sigma_n \leq \beta$. All the relevant quantities are finite here, since a global bound on s_n also bounds each average.

If instead $\beta = +\infty$, the desired upper bound is simply $\limsup \sigma_n \leq +\infty$, which always holds. No finite-tail estimate involving an infinite supremum is required.

(a), lower bound: retain a uniformly positive tail.

If $\alpha = 0$, nonnegativity already gives $\liminf \sigma_n \geq 0 = \alpha$. If $\alpha > 0$, choose an arbitrary real c with $0 < c < \alpha$. For $\alpha = +\infty$, take any $c > 0$. There is N such that $s_j \geq c$ for all $j > N$, by the definition of the limit inferior. For $n > N$, discard the nonnegative initial sum and estimate the tail:

$$\sigma_n = \frac{P_N + s_{N+1} + \cdots + s_n}{n} \geq \frac{(n-N)c}{n} = c\left(1 - \frac{N}{n}\right).$$

The expression on the right tends to c, since N is fixed. Exercise 12.1 gives $\liminf \sigma_n \geq c$. When α is finite, letting c range over all positive numbers below α forces $\liminf \sigma_n \geq \alpha$: otherwise choose c between these two quantities. When $\alpha = +\infty$, the same inequality for every $c > 0$ forces $\liminf \sigma_n = +\infty$. Combining the three inequalities proves

$$\boxed{\liminf s_n \leq \liminf \sigma_n \leq \limsup \sigma_n \leq \limsup s_n.}$$

(b) Convergence passes to the averages.

Suppose $s_n \to L$. Nonnegativity means that either $L \geq 0$ is finite or $L = +\infty$; a limit of $-\infty$ is impossible. Then $\liminf s_n = \limsup s_n = L$, and part (a) gives

$$L \leq \liminf \sigma_n \leq \limsup \sigma_n \leq L.$$

Thus both middle quantities equal L. Theorem 10.7, which characterizes convergence by equality of the limit inferior and limit superior, implies $\sigma_n \to L$. If $L = +\infty$, this statement means that the averages also tend to $+\infty$.

(c) Convergent averages need not imply convergent terms.

Take $s_n = 1 + (-1)^n$. Its terms alternate $0, 2, 0, 2, \ldots$, and are nonnegative. The odd subsequence is constantly 0 and the even subsequence is constantly 2, so (s_n) has no limit.

For $n = 2k$, there are k zeros and k twos, so $\sigma_{2k} = 2k/(2k) = 1$. For $n = 2k - 1$, there are k zeros and $k - 1$ twos, so

$$\sigma_{2k-1} = \frac{2(k-1)}{2k-1} = 1 - \frac{1}{2k-1}.$$

In both cases $|\sigma_n - 1| \leq 1/n$. Given $\varepsilon > 0$, choose $N > 1/\varepsilon$; then $n > N$ implies $|\sigma_n - 1| \leq 1/n < \varepsilon$. Thus $\sigma_n \to 1$.

Conclusion. The averages satisfy the four-term bounds and inherit every limit of a nonnegative sequence. The converse fails for alternating zeros and twos.

12.13 Exercise 12.13: Limit inferior and superior as eventual bounds

What we must show. For a bounded real sequence define

$$A = \{a \in \mathbb{R} : \{n \in \mathbb{N} : s_n < a\} \text{ is finite}\},$$
$$B = \{b \in \mathbb{R} : \{n \in \mathbb{N} : s_n > b\} \text{ is finite}\}.$$

Prove $\sup A = \liminf s_n$ and $\inf B = \limsup s_n$.

Plan. A finite set of exceptional indices can be placed before a single cutoff. Thus A consists of eventual lower bounds and B of eventual upper bounds. The boundary value itself need not belong to the respective set.

Check existence of the real suprema and infima.

Choose $C \geq 0$ with $-C \leq s_n \leq C$ for every n. For $a = -C$, no term satisfies $s_n < a$, so $-C \in A$. If $a > C$, every index satisfies $s_n < a$, so $a \notin A$. Thus A is nonempty and bounded above by C. Similarly, $C \in B$, and every $b < -C$ lies outside B, since $s_n > b$ for every n. Hence B is nonempty and bounded below by $-C$. Completeness gives finite real values $\sup A$ and $\inf B$. Write

$$\alpha = \liminf s_n, \quad \beta = \limsup s_n, \quad L_N = \inf\{s_n : n > N\}, \quad U_N = \sup\{s_n : n > N\}.$$

All these numbers are finite, and $L_N \to \alpha$, $U_N \to \beta$.

Every eventual lower bound is at most α.

If $a \in A$, its finitely many exceptional indices lie at most at some index N_0; if there are none, take $N_0 = 1$. Then $s_n \geq a$ whenever $n > N_0$. For $N \geq N_0$, it follows that $L_N \geq a$, and passage to the limit gives $\alpha \geq a$. Thus α is an upper bound of A, so $\sup A \leq \alpha$.

No number below α can be an upper bound of A.

If $a < \alpha$, convergence $L_N \to \alpha$ implies $L_N > a$ for some N. For every $n > N$, we have $s_n \geq L_N > a$. Consequently the indices with $s_n < a$ are contained in $\{1, \ldots, N\}$, a finite set; hence $a \in A$. Now fix any $d < \alpha$ and take $a = (d + \alpha)/2$. Then $d < a < \alpha$, so $a \in A$ and $a > d$. Therefore d fails to be an upper bound of A. This proves that α is the least upper bound: $\sup A = \alpha$.

Repeat the two directions for eventual upper bounds.

If $b \in B$, there is N_0 such that $s_n \leq b$ whenever $n > N_0$. Then $U_N \leq b$ for all $N \geq N_0$; taking limits gives $\beta \leq b$. Thus β is a lower bound of B.

If $b > \beta$, convergence $U_N \to \beta$ gives some N with $U_N < b$. For every $n > N$, we have $s_n \leq U_N < b$, so the indices satisfying $s_n > b$ form a finite set. Thus $b \in B$. If $d > \beta$, the number $b = (\beta + d)/2$ satisfies $\beta < b < d$ and belongs to B, so d cannot be a lower bound of B. Hence β is its greatest lower bound: $\inf B = \beta$.

Why the boundary is not asserted to lie in the set.

For $s_n = -1/n$, one has $\alpha = 0$, but every $s_n < 0$, so $0 \notin A$. For $s_n = 1/n$, one has $\beta = 0$, but every $s_n > 0$, so $0 \notin B$. Suprema and infima need not be attained.

Conclusion. The least eventual upper bound is $\limsup s_n$, and the greatest eventual lower bound is $\liminf s_n$, in the precise supremum/infimum sense above.

12.14 Exercise 12.14: The growth of the geometric mean of the first integers

What we must show. Calculate the limits of $(n!)^{1/n}$ and $(n!)^{1/n}/n$.

Plan. Apply the ratio–root corollary to two different positive sequences. For the second limit we also justify the elementary sequence that defines e, without using Stirling's formula.

(a) Apply the ratio–root corollary to the factorials.

Let $v_n = n!$, which is positive. Cancelling the common product $1 \cdot 2 \cdots n$ gives

$$\frac{v_{n+1}}{v_n} = \frac{(n+1)!}{n!} = n+1 \longrightarrow +\infty.$$

Corollary 12.3 states that a limit of the absolute successive ratios is also the limit of the nth roots, including the value $+\infty$. Therefore

$$\boxed{\lim_{n\to\infty} (n!)^{1/n} = +\infty.}$$

(b), preparation: justify the sequence defining e.

Set $b_n = (1 + 1/n)^n$. The standard constant e is the limit of this sequence, as described in the text's Section 7, Example 1(e). For completeness we verify that this limit exists and is positive. By the binomial theorem,

$$b_n = \sum_{k=0}^{n} \binom{n}{k} \frac{1}{n^k} = \sum_{k=0}^{n} \frac{1}{k!} \prod_{j=0}^{k-1} \left(1 - \frac{j}{n}\right).$$

For $k = 0$ the product is empty and equals 1. For fixed $k \le n$, each displayed factor is nonnegative and does not decrease when n is replaced by $n+1$. Thus each of the old summands does not decrease. The extra summand with $k = n+1$ in the expression for b_{n+1} is positive, so $b_{n+1} > b_n$.

Every factor is at most 1. For $k \ge 2$, the $k-1$ factors $2, 3, \ldots, k$ in $k!$ are each at least 2, whence $k! \ge 2^{k-1}$. It follows that

$$b_n \le \sum_{k=0}^{n} \frac{1}{k!} \le 2 + \sum_{k=2}^{n} \frac{1}{2^{k-1}} < 3.$$

For $n = 1$ the last sum is empty, and the same bound holds. Since $b_1 = 2$, the monotone sequence theorem now gives $b_n \to e$ for a finite number $e \ge 2 > 0$.

(b), normalization: put the denominator inside the root.

Define $u_n = n!/n^n > 0$. Its positive nth root is

$$u_n^{1/n} = \frac{(n!)^{1/n}}{n},$$

because the nth power of the expression on the right is $n!/n^n$ and the positive nth root is unique. The successive ratio simplifies one factor at a time:

$$\begin{aligned} \frac{u_{n+1}}{u_n} &= \frac{(n+1)!}{(n+1)^{n+1}} \frac{n^n}{n!} \\ &= \frac{(n+1)n^n}{(n+1)^{n+1}} = \left(\frac{n}{n+1}\right)^n = \frac{1}{(1+1/n)^n} = \frac{1}{b_n}. \end{aligned}$$

Since $b_n \to e \neq 0$, the reciprocal limit law gives $u_{n+1}/u_n \to 1/e$. Corollary 12.3 therefore gives $u_n^{1/n} \to 1/e$, which is exactly

$$\boxed{\lim_{n\to\infty} \frac{(n!)^{1/n}}{n} = \frac{1}{e}.}$$

Conclusion. The unnormalized geometric mean tends to $+\infty$; after division by n, it tends to $1/e$.